POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6043A | Semester VI | Theory | 4 Credits

Fundamentals of Hybrid & Electric Vehicles

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6043A as Fundamentals of Hybrid & Electric Vehicles in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Introduction to Hybrid and Electric Vehicles course and polytechnic EV technology curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. EV designs, hybrid architectures, BMS logic and charging systems must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support EV/HEV architectures, BMS, propulsion systems, and charging infrastructure. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Fundamentals of Hybrid & Electric Vehicles studies the principles, architectures and control of electrified transportation systems. It develops the ability to analyze hybrid and electric vehicle (EV) powertrains, evaluate battery management systems (BMS), understand electric propulsion and regenerative braking, and coordinate EV charging solutions while respecting the technical and safety boundaries of electronic and automotive engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electrical engineering, power electronics, control systems and vehicle dynamics basics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the various architectures and social impacts of hybrid and electric vehicle (EV) systems.
2. Analyze the operation, management and safety of energy storage systems in electrified vehicles.
3. Evaluate the performance and control of electric motors and power electronic converters for EV propulsion.
4. Explain the systematic design, regenerative braking and charging infrastructure for modern EVs.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO □□□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the architectures and components of hybrid and electric vehicle systems.	Understand
CO2	Analyze the management and safety of EV energy storage and battery systems.	Apply
CO3	Analyze the characteristics and control of electric propulsion and motor drives.	Apply
CO4	Explain regenerative braking, vehicle dynamics and EV charging standards.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Hybrid and EV Architectures	History and social impact of EVs; Hybrid Electric Vehicle (HEV) architectures: Series, Parallel, Series-Parallel (Power-split); Plug-in Hybrid Electric Vehicles (PHEV); Battery Electric Vehicles (BEV).	Approx. 14 hours	CO1	Module 1
2	Energy Storage and Battery Management	Battery technologies: Li-ion, Ni-MH, Lead-acid; Cell balancing and thermal management; SoC and SoH estimation; Battery Management Systems (BMS); Supercapacitors and Fuel cells basics.	Approx. 16 hours	CO2	Module 2
3	Electric Propulsion and Control	Electric motors for EVs: BLDC, Induction, PMSM; Power electronic converters: Inverters, DC-DC converters; Motor control techniques: PWM, Vector control; Regenerative braking principles.	Approx. 16 hours	CO3	Module 3
4	Vehicle Dynamics and Infrastructure	Tractive effort and vehicle performance; Rolling resistance, Aerodynamic drag, Grading resistance; EV charging levels (1, 2, 3) and standards (CCS, CHAdeMO); Grid impact of EV charging.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Hybrid and EV Architectures

History and social impact of EVs; Hybrid Electric Vehicle (HEV) architectures: Series, Parallel, Series-Parallel (Power-split); Plug-in Hybrid Electric Vehicles (PHEV); Battery Electric Vehicles (BEV).

CO1 · Approx. 14 hours

Energy Storage and Battery Management

Battery technologies: Li-ion, Ni-MH, Lead-acid; Cell balancing and thermal management; SoC and SoH estimation; Battery Management Systems (BMS); Supercapacitors and Fuel cells basics.

CO2 · Approx. 16 hours

Electric Propulsion and Control

Electric motors for EVs: BLDC, Induction, PMSM; Power electronic converters: Inverters, DC-DC converters; Motor control techniques: PWM, Vector control; Regenerative braking principles.

CO3 · Approx. 16 hours

Vehicle Dynamics and Infrastructure

Tractive effort and vehicle performance; Rolling resistance, Aerodynamic drag, Grading resistance; EV charging levels (1, 2, 3) and standards (CCS, CHAdeMO); Grid impact of EV charging.

CO4 · Approx. 14 hours

MODULE 1

Hybrid and EV Architectures

History and social impact of EVs; Hybrid Electric Vehicle (HEV) architectures: Series, Parallel, Series-Parallel (Power-split); Plug-in Hybrid Electric Vehicles (PHEV); Battery Electric Vehicles (BEV).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

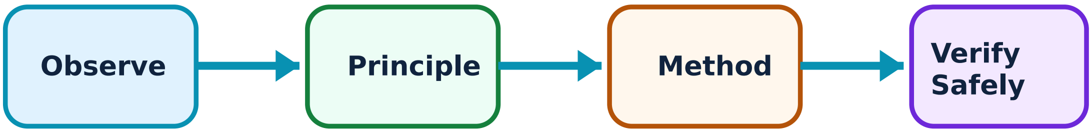
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Hybrid and EV Architectures: identify the system, state its principle, follow the correct method and verify the result safely.

1. Introduction to Electrified Vehicles−

Electrified vehicles reduce greenhouse gas emissions and reliance on fossil fuels. HEVs combine an internal combustion engine (ICE) with an electric motor. BEVs rely solely on batteries. PHEVs can be charged from the grid and have a larger electric-only range.

Study and application method

Compare 'BEV', 'HEV', and 'PHEV' in a conceptual table; explain the social benefits of vehicle electrification.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'BEV', 'HEV', and 'PHEV' in a conceptual table; explain the social benefits of vehicle electrification.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: EV adoption depends on grid capacity and infrastructure. Do not assume universal EV feasibility without authorised local power grid and charging station analysis.

2. Powertrain Architectures−

In a Series hybrid, the engine drives a generator to charge the battery or power the motor. In a Parallel hybrid, both the engine and motor can drive the wheels. Series-Parallel hybrids use a power-split device to allow both modes, optimizing efficiency.

Study and application method

Sketch the block diagram of a 'Parallel Hybrid' powertrain; explain the function of a 'Power-split Device'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of a 'Parallel Hybrid' powertrain; explain the function of a 'Power-split Device'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Hybrid powertrains involve complex mechanical and electrical coupling. All powertrain designs must follow authorised control strategies and safety interlocking.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Energy Storage and Battery Management

Battery technologies: Li-ion, Ni-MH, Lead-acid; Cell balancing and thermal management; SoC and SoH estimation; Battery Management Systems (BMS); Supercapacitors and Fuel cells basics.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

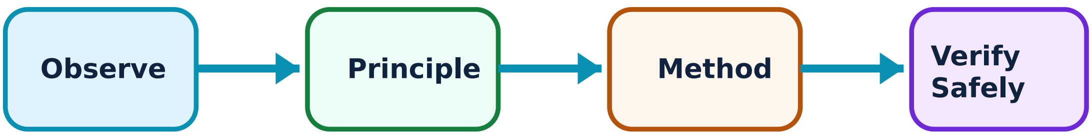
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Storage and Battery Management: identify the system, state its principle, follow the correct method and verify the result safely.

1. Battery Technology and Management−

Lithium-ion batteries are preferred for their high energy density. A Battery Management System (BMS) monitors voltage, current, and temperature to ensure safety. It estimates the State of Charge (SoC) and State of Health (SoH) and performs cell balancing to maximize battery life.

Study and application method

Define 'State of Charge' (SoC); explain why 'Cell Balancing' is necessary in a multi-cell battery pack.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'State of Charge' (SoC); explain why 'Cell Balancing' is necessary in a multi-cell battery pack.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Lithium batteries can experience thermal runaway if mistreated. Do not operate battery packs without authorised BMS protection and thermal management systems.

2. Alternative Storage and Safety−

Supercapacitors provide high power for short bursts, while Fuel cells generate electricity from hydrogen. Safety in EVs involves high-voltage isolation, interlocks, and emergency disconnects to protect passengers and first responders.

Study and application method

Describe the role of 'Supercapacitors' in regenerative braking; explain the importance of 'High-voltage Isolation' in EVs.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of 'Supercapacitors' in regenerative braking; explain the importance of 'High-voltage Isolation' in EVs.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High-voltage systems (typically 400V-800V) are lethal. Do not perform maintenance on EV electrical systems without authorised high-voltage safety training and PPE.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Electric Propulsion and Control

Electric motors for EVs: BLDC, Induction, PMSM; Power electronic converters: Inverters, DC-DC converters; Motor control techniques: PWM, Vector control; Regenerative braking principles.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

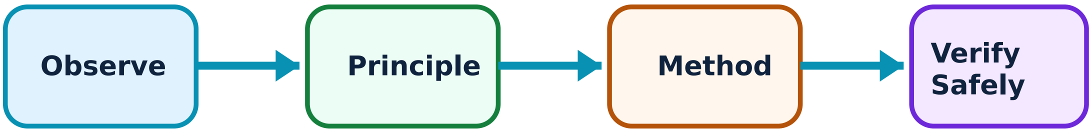
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electric Propulsion and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. EV Motors and Converters−

PMSM and Induction motors are widely used for their efficiency and power density. Inverters convert battery DC into AC for the motor. DC-DC converters are used to step down the high battery voltage for low-voltage auxiliary systems (lights, infotainment).

Study and application method

Compare 'BLDC' and 'PMSM' motors for EV applications; explain the role of a 'Bi-directional DC-DC Converter' in a hybrid system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'BLDC' and 'PMSM' motors for EV applications; explain the role of a 'Bi-directional DC-DC Converter' in a hybrid system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inverters generate significant electromagnetic interference (EMI). All power electronic designs must follow authorised EMI/EMC standards and shielding protocols.

2. Propulsion Control and Braking−

Vector control (Field Oriented Control) provides precise torque and speed regulation. Regenerative braking allows the motor to act as a generator during deceleration, converting kinetic energy back into electrical energy to charge the battery.

Study and application method

Describe the principle of 'Regenerative Braking' conceptually; explain how 'PWM' is used to control motor speed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the principle of 'Regenerative Braking' conceptually; explain how 'PWM' is used to control motor speed.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Regenerative braking is limited by battery SoC and temperature. All braking systems must include authorised mechanical backups that are fully independent of the electrical system.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Vehicle Dynamics and Infrastructure

Tractive effort and vehicle performance; Rolling resistance, Aerodynamic drag, Grading resistance; EV charging levels (1, 2, 3) and standards (CCS, CHAdeMO); Grid impact of EV charging.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

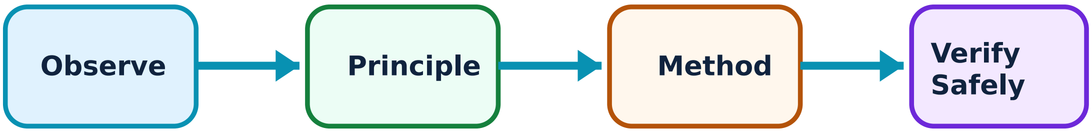
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Vehicle Dynamics and Infrastructure: identify the system, state its principle, follow the correct method and verify the result safely.

1. Vehicle Dynamics−

Tractive effort must overcome rolling resistance (tire-road interaction), aerodynamic drag (air resistance), and grading resistance (on hills). The energy required per km (Wh/km) determines the vehicle's range for a given battery capacity.

Study and application method

List the three main forces opposing vehicle motion; calculate the 'Tractive Effort' required for a given acceleration conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the three main forces opposing vehicle motion; calculate the 'Tractive Effort' required for a given acceleration conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Range estimation is highly sensitive to driving style and environment. Do not provide range guarantees without authorised testing under standardized drive cycles (e.g., WLTP).

2. Charging Infrastructure−

Charging levels range from slow AC (Level 1) to ultra-fast DC (Level 3). Standards like CCS and CHAdeMO ensure interoperability. Large-scale EV charging requires smart grid management to prevent transformer overloading and to optimize energy costs.

Study and application method

Differentiate between 'Level 2' and 'Level 3' charging; explain the concept of 'V2G' (Vehicle-to-Grid) conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Level 2' and 'Level 3' charging; explain the concept of 'V2G' (Vehicle-to-Grid) conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper charging can degrade the battery or cause grid instability. All charging installations must follow authorised electrical codes and utility interconnection standards.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Hybrid and EV Architectures

Use the concept flow to revise observation, principle, method and verification.

Module 2: Energy Storage and Battery Management

Use the concept flow to revise observation, principle, method and verification.

Module 3: Electric Propulsion and Control

Use the concept flow to revise observation, principle, method and verification.

Module 4: Vehicle Dynamics and Infrastructure

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Series and Parallel hybrid architectures in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the energy flow in an EV during 'Motoring' and 'Regenerative Braking'.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a Battery Management System (BMS) showing key sensors and controllers.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the energy required to charge a 40kWh battery from 20% to 80% SoC conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a local EV charging station and note its connector types and power rating if possible.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Hybrid and EV Architectures

Explain Introduction to Electrified Vehicles and state one correct method, application or precaution.−

Electrified vehicles reduce greenhouse gas emissions and reliance on fossil fuels. HEVs combine an internal combustion engine (ICE) with an electric motor. BEVs rely solely on batteries. PHEVs can be charged from the grid and have a larger electric-only range.

Answer framework: Compare 'BEV', 'HEV', and 'PHEV' in a conceptual table; explain the social benefits of vehicle electrification.

Important point: EV adoption depends on grid capacity and infrastructure. Do not assume universal EV feasibility without authorised local power grid and charging station analysis.

Explain Powertrain Architectures and state one correct method, application or precaution.−

In a Series hybrid, the engine drives a generator to charge the battery or power the motor. In a Parallel hybrid, both the engine and motor can drive the wheels. Series-Parallel hybrids use a power-split device to allow both modes, optimizing efficiency.

Answer framework: Sketch the block diagram of a 'Parallel Hybrid' powertrain; explain the function of a 'Power-split Device'.

Important point: Hybrid powertrains involve complex mechanical and electrical coupling. All powertrain designs must follow authorised control strategies and safety interlocking.

Module 2 — Energy Storage and Battery Management

Explain Battery Technology and Management and state one correct method, application or precaution.−

Lithium-ion batteries are preferred for their high energy density. A Battery Management System (BMS) monitors voltage, current, and temperature to ensure safety. It estimates the State of Charge (SoC) and State of Health (SoH) and performs cell balancing to maximize battery life.

Answer framework: Define 'State of Charge' (SoC); explain why 'Cell Balancing' is necessary in a multi-cell battery pack.

Important point: Lithium batteries can experience thermal runaway if mistreated. Do not operate battery packs without authorised BMS protection and thermal management systems.

Explain Alternative Storage and Safety and state one correct method, application or

precaution.−

Supercapacitors provide high power for short bursts, while Fuel cells generate electricity from hydrogen. Safety in EVs involves high-voltage isolation, interlocks, and emergency disconnects to protect passengers and first responders.

Answer framework: Describe the role of 'Supercapacitors' in regenerative braking; explain the importance of 'High-voltage Isolation' in EVs.

Important point: High-voltage systems (typically 400V-800V) are lethal. Do not perform maintenance on EV electrical systems without authorised high-voltage safety training and PPE.

Module 3 — Electric Propulsion and Control

Explain EV Motors and Converters and state one correct method, application or precaution.−

PMSM and Induction motors are widely used for their efficiency and power density. Inverters convert battery DC into AC for the motor. DC-DC converters are used to step down the high battery voltage for low-voltage auxiliary systems (lights, infotainment).

Answer framework: Compare 'BLDC' and 'PMSM' motors for EV applications; explain the role of a 'Bi-directional DC-DC Converter' in a hybrid system.

Important point: Inverters generate significant electromagnetic interference (EMI). All power electronic designs must follow authorised EMI/EMC standards and shielding protocols.

Explain Propulsion Control and Braking and state one correct method, application or precaution.−

Vector control (Field Oriented Control) provides precise torque and speed regulation. Regenerative braking allows the motor to act as a generator during deceleration, converting kinetic energy back into electrical energy to charge the battery.

Answer framework: Describe the principle of 'Regenerative Braking' conceptually; explain how 'PWM' is used to control motor speed.

Important point: Regenerative braking is limited by battery SoC and temperature. All braking systems must include authorised mechanical backups that are fully independent of the electrical system.

Module 4 — Vehicle Dynamics and Infrastructure

Explain Vehicle Dynamics and state one correct method, application or precaution.−

Tractive effort must overcome rolling resistance (tire-road interaction), aerodynamic drag (air resistance), and grading resistance (on hills). The energy required per km (Wh/km) determines the vehicle's range for a given battery capacity.

Answer framework: List the three main forces opposing vehicle motion; calculate the 'Tractive Effort' required for a given acceleration conceptually.

Important point: Range estimation is highly sensitive to driving style and environment. Do not provide range guarantees without authorised testing under standardized drive cycles (e.g., WLTP).

Explain Charging Infrastructure and state one correct method, application or precaution.−

Charging levels range from slow AC (Level 1) to ultra-fast DC (Level 3). Standards like CCS and CHAdeMO ensure interoperability. Large-scale EV charging requires smart grid management to prevent transformer overloading and to optimize energy costs.

Answer framework: Differentiate between 'Level 2' and 'Level 3' charging; explain the concept of 'V2G' (Vehicle-to-Grid) conceptually.

Important point: Improper charging can degrade the battery or cause grid instability. All charging installations must follow authorised electrical codes and utility interconnection standards.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Hybrid and EV Architectures.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Energy Storage and Battery Management.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Electric Propulsion and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Vehicle Dynamics and Infrastructure.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: History and social impact of EVs; Hybrid Electric Vehicle (HEV) architectures: Series, Parallel, Series-Parallel (Power-split); Plug-in Hybrid Electric Vehicles (PHEV); Battery Electric Vehicles (BEV)..
2. Develop a complete answer or practical plan for: Battery technologies: Li-ion, Ni-MH, Lead-acid; Cell balancing and thermal management; SoC and SoH estimation; Battery Management Systems (BMS); Supercapacitors and Fuel cells basics..
3. Develop a complete answer or practical plan for: Electric motors for EVs: BLDC, Induction, PMSM; Power electronic converters: Inverters, DC-DC converters; Motor control techniques: PWM, Vector control; Regenerative braking principles..
4. Develop a complete answer or practical plan for: Tractive effort and vehicle performance; Rolling resistance, Aerodynamic drag, Grading resistance; EV charging levels (1, 2, 3) and standards (CCS, CHAdeMO); Grid impact of EV charging..

LAST-MILE REVISION

Quick Revision

Module 1: Hybrid and EV Architectures

History and social impact of EVs; Hybrid Electric Vehicle (HEV) architectures: Series, Parallel, Series-Parallel (Power-split); Plug-in Hybrid Electric Vehicles (PHEV); Battery Electric Vehicles (BEV).

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Energy Storage and Battery Management

Battery technologies: Li-ion, Ni-MH, Lead-acid; Cell balancing and thermal management; SoC and SoH estimation; Battery Management Systems (BMS); Supercapacitors and Fuel cells basics.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Electric Propulsion and Control

Electric motors for EVs: BLDC, Induction, PMSM; Power electronic converters: Inverters, DC-DC converters; Motor control techniques: PWM, Vector control; Regenerative braking principles.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Vehicle Dynamics and Infrastructure

Tractive effort and vehicle performance; Rolling resistance, Aerodynamic drag, Grading resistance; EV charging levels (1, 2, 3) and standards (CCS, CHAdeMO); Grid impact of EV charging.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Introduction to Electrified Vehicles	Electrified vehicles reduce greenhouse gas emissions and reliance on fossil fuels.
Powertrain Architectures	In a Series hybrid, the engine drives a generator to charge the battery or power the motor.
Battery Technology and Management	Lithium-ion batteries are preferred for their high energy density.
Alternative Storage and Safety	Supercapacitors provide high power for short bursts, while Fuel cells generate electricity from hydrogen.
EV Motors and Converters	PMSM and Induction motors are widely used for their efficiency and power density.
Propulsion Control and Braking	Vector control (Field Oriented Control) provides precise torque and speed regulation.
Vehicle Dynamics	Tractive effort must overcome rolling resistance (tire-road interaction), aerodynamic drag (air resistance), and grading resistance (on hills).
Charging Infrastructure	Charging levels range from slow AC (Level 1) to ultra-fast DC (Level 3).

LEARNING RESOURCES

References

Recommended hybrid and EV references

1. Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals.
2. Mehrdad Ehsani, Yimin Gao and Ali Emadi, Modern Electric, Hybrid Electric, and Fuel Cell Vehicles.
3. James Larminie and John Lowry, Electric Vehicle Technology Explained.
4. L. Guzzella and A. Sciarretta, Vehicle Propulsion Systems: Introduction to Modeling and Optimization.

Approved public hybrid and EV sources

- https://nptel.ac.in/courses/108106170
- https://onlinecourses.nptel.ac.in/noc25_ee33/preview

These sources provide the verified study basis while official Course 6043A detail is unavailable; they do not replace official automotive designs, authorised safety protocols, certified equipment specifications or authorised transport plans.